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The Pitch-Yaw Wave Number (rev.3)

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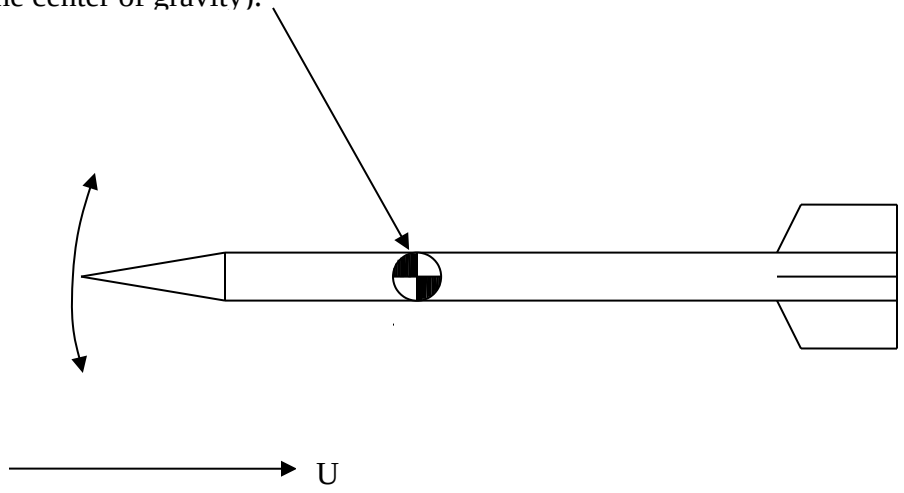
Introduction

Most of you have heard me speak about the pitch/yaw wave number, and your eyes just glaze over when you hear that phrase. What sort of techno-mumbo-jumbo can that be? you're thinking. Here's the answer:

Discussion

First, our rockets are all more or less symmetric around their roll axes. Their pitch moments of inertia are the same as their yaw moments of inertia. And, the aerodynamic torques about the pitch axis are the same as about their yaw axis.

Now, imagine a rocket model suspended horizontally by a pitch axle passing through the center of gravity. The rocket model is free to rotate about the pitch axis (passing through the center of gravity).



Now, imagine the model in a wind tunnel. Further, suppose you reached in and tweaked the nose such that the model would freely oscillate. Then, the differential equation describing the oscillations is

$$Id^2\alpha/dt^2 = \tfrac{1}{2}\rho S d U^2 C_{M\alpha}\alpha,$$

where I = Pitch moment of inertia,
L = Pitch/yaw wave length,
ℓ = Rocket length,
t = Time,
ρ = Atmospheric mass density,
S = Reference area for aerodynamic coefficients, body cross section area
d = Reference length for aerodynamic moment coefficients, body diameter
U = Airspeed
$C_{M\alpha}$ = Slope of pitch moment coefficient with respect to angle of attack, and
α = Angle of attack.

To solve this differential equation, substitute

$$\alpha = \sin \omega_N t$$

After a bit of math, you will find that

$$\omega_N^2 = -\rho S d U^2 C_{M\alpha}/2I$$

The quantity ω_N is called the pitch natural frequency, and, as already noted, due to symmetry about the roll axis, it is the same as the yaw natural frequency.

The pitch period P is the time required for one complete pitch oscillation…that is, the time for α to go through 2π radians, or

$$\omega_N P = 2\pi$$

$$P = 2\pi/\omega_N$$

But, during one oscillation period, the rocket will have moved through the air a distance $L = UP$. This distance L is the pitch wavelength. After a little more math you'll find that

$$L = 2\pi\sqrt{2I/-\rho S d C_{M\alpha}}$$

Don't worry about the minus sign; $C_{M\alpha}$ is a negative number for a stable rocket.

The pitch wave number λ_P is a kind of frequency in the spatial domain instead of the time domain. Instead of radians per second its units are radians per foot (or meter). Numerically it's just

$$\lambda_P = \sqrt{-\rho S d C_{M\alpha}/2I}$$

Finally, and most remarkably, the same formula holds when U is not a constant, but is increasing under constant acceleration. Trust me. Even better, check it out yourself!

Finally, consider how the pitch wavelength scales with rocket size. Imagine two rockets, one scaled up from the other. In the equation for wavelength some parameters are not scale dependent. These are density ρ , and $C_{M\alpha}$. However, Sd will be proportional to ℓ^3 and I is proportional to ℓ^5 because the mass is proportional to ℓ^3 and the radius of gyration is proportional to ℓ. Putting this all together,

$$L \propto \ell .$$

In other words, bigger rockets have longer pitch/yaw wavelengths.